

English
Medium



EDU TERIA

72nd BPSC

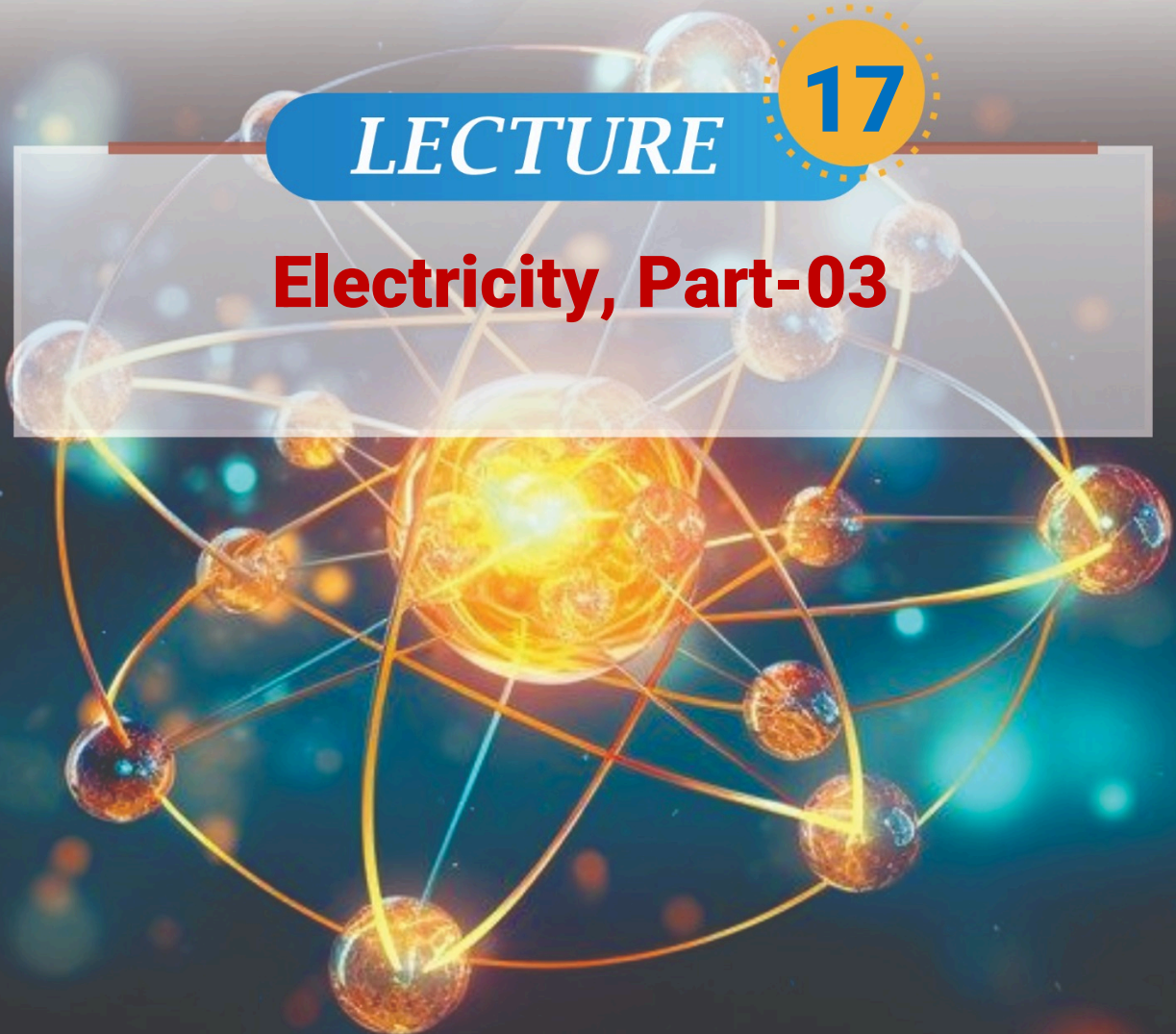
Prelims Online Batch

PHYSICS

LECTURE

17

Electricity, Part-03



**Electricity, Part-03****Electrical Power**

Electrical power is the rate at which electrical energy is transferred or used in a circuit. In simple terms, it tells you how much electricity is being consumed or produced at any moment.

Formula of Power

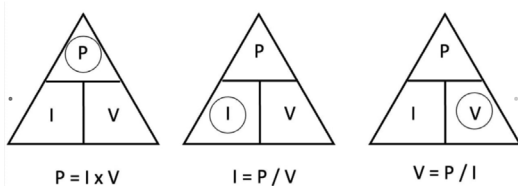
$$P = I^2 R$$

$$P = \frac{V}{R} \times V = \frac{V^2}{R}$$

$$P = VI = I^2 R = \frac{V^2}{R}$$

- V = Potential difference
- I = Electric current
- R = Resistance
- P = Electrical Power

$$\therefore P = VI = I^2 R = \frac{V^2}{R}$$



For a given circuit of R—constant

$P \propto V^2$ If voltage remain constant in circuit:

$$P \propto I^2 \quad P \propto I$$

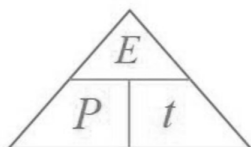
$$\frac{P_1}{P_2} = \frac{I_1}{I_2}$$

Electrical Energy

Electrical energy is the total work done by an electric current in a circuit over a given time, i.e., the amount of electrical power consumed or supplied during that time.

$$E = Pt$$

$$P = \frac{E}{t}$$



E = Energy in Joules (J)

P = Power in Watts (W)

t = Time in Seconds (s)

$$E = Pt \quad \text{power} = \frac{\text{Energy}}{\text{Time}}$$

Commercial unit : kWh

1 unit = 1 kWh

Electrical Energy = Power $\times$ Time, Unit= Watt second

Commercial Unit = Kilo watt hour

1 kilowatt-hour (kWh) = 3.6×10^6 Joules.

Effect of current**1. Heating Effect of Current**

When current passes through a conductor, electrical energy converts into heat.

Based on Joule's Law of Heating

Heat produced:

$$H = I^2 R t$$

Used in heaters, irons

$$\text{Heat (h)} = I^2 R t$$

Examples:

Electric heater, Iron, Filament bulb, Used in appliances where heat is required.

2. Magnetic Effect of Current

A current-carrying conductor produces a magnetic field around it.

Explained by Oersted's Experiment

Examples:

Electromagnets, Electric motor, Relay, Basis of modern electrical machines.

3. Chemical Effect of Current

Electric current causes chemical changes in a conducting liquid (electrolyte).

Studied in electrolysis

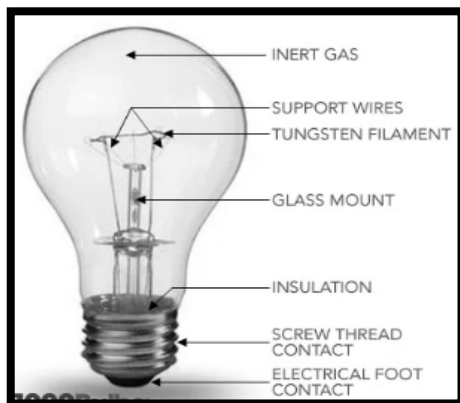
Examples:

Electroplating, Battery charging, Extraction of metals, Important in industry and chemistry.

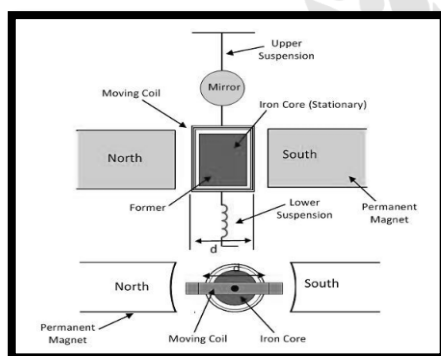
Application of Heating effect of current:—

- Electrical fuse**—Ideal electrical fuse is made up of tin and lead (Sn+Pb)
It is connected in series with main electrical switchboard.
- Electrical bulb**—Works on heating effect of current.

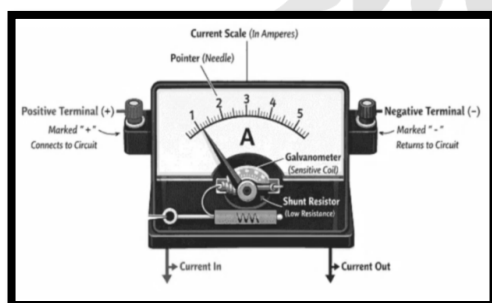


**Electrical Instrument—**

1. **Galvanometer**—Arrangement which is used to detect presence of current in circuit.

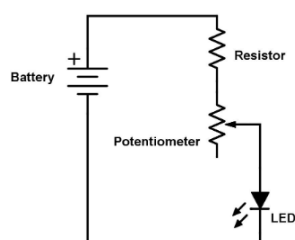


2. **Ammeter**—Modified galvanometer which is used to measure amount of flow of current electrical circuit. It is connected in series.



3. **Voltmeter**—Used to measure electrical potential of circuit. It is connected in parallel.
4. **Potentiometer**—Highly sensitive instrument to measure emf.

Emf = Electrometer force or potential difference or voltage.



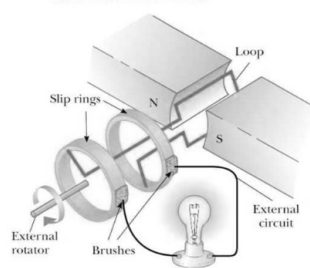
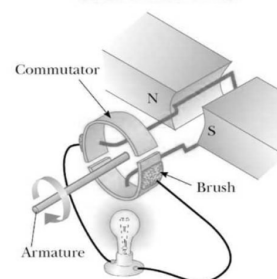
5. **Voltmeter**—Instrument used to measure charge flow in electrical circuit.

6. Wattmeter

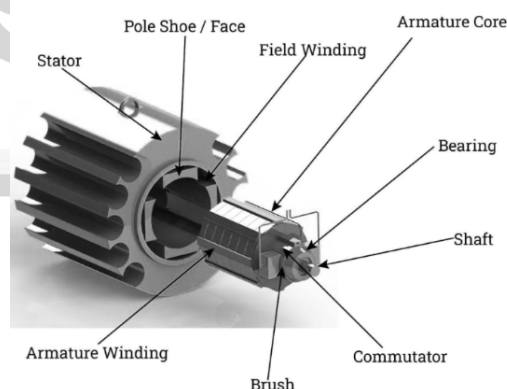
- Function: Measures electrical power
- Used in AC/DC circuits
- Has two coils:
 1. Current coil
 2. Potential (voltage) coil

Electrical Devices and Energy Conversion**Dynamo / Generator**

- Converts mechanical energy → electrical energy. Works on electromagnetic induction.
- AC generator uses slip rings.
- DC generator uses a commutator

Generator AC**Generator DC****Electric Motor**

- Converts electrical energy → mechanical energy (motion).
- Also works on electromagnetic induction.
- Low-voltage motors tend to burn out because they draw more current ($I = V/R$).

DC Motor Diagram**Energy Conversion Efficiency**

- In an ideal motor converting electrical energy directly to motion, there is no heat loss. (Note: Real-world applications always have some heat loss).



**Batteries**

- **General Principle:** Converts stored chemical energy → electrical energy.
- **Components:** Positive electrode, negative electrode, and an electrolyte.

Types:**Lead-Acid Battery (e.g., Car Battery):**

- **Voltage:** 12V (from six 2V cells).
- **Electrodes:** Lead and Lead dioxide.
- **Electrolyte:** Sulfuric acid and water (~35% acid).
- **Capacity:** Expressed in Ampere-hour (Ah).

Nickel-Cadmium (Ni-Cd) Battery:

- Rechargeable.
- **Electrodes:** Nickel oxide hydroxide and Metallic cadmium.
- **Electrolyte:** Alkaline Potassium hydroxide.
- **Uses:** Calculators, cordless appliances, transistors, portable tools.

Transformers and Current Conversion:**Transformer**

- **Purpose:** Steps up or steps down AC voltage.
- **Principle:** Electromagnetic induction; only works with AC, not DC.
- **Application:** A mobile phone charger is a step-down transformer (e.g., 220V AC to 9V DC).

AC/DC Conversion

- **Rectifier:** Converts AC to DC (needed for charging batteries).
- **Inverter:** Converts DC to AC.
- **Frequency in India:** The AC system operates at 50 Hz.

Lighting Technologies

- **Filament:** Made of Tungsten (high melting point: 3422°C).
- **Operating Temperature:** 2000°C to 2500°C.
- **Gas Filling:** Inert gases like Nitrogen or Argon to prevent oxidation.

Incandescent Bulbs**Reasons for Short Lifespan:**

- Filament wire is not perfectly uniform.

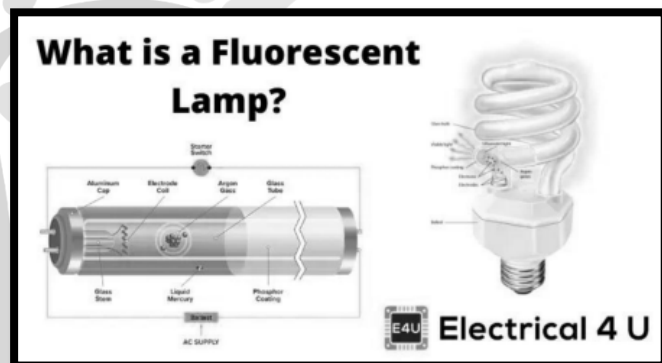
- Bulb cannot be completely evacuated of air.
- Supporting wires can melt. BPSC PRELIMS
- **Very inefficient:** only 5-10% of energy is converted to light; the rest is heat.

Halogen Lamps

- The filament is an alloy of Tungsten and Sodium.

Fluorescent Lamps (Tube Lights & CFLs)

- **CFL:** Stands for Compact Fluorescent Lamp.
- **Gas:** Contains Mercury vapor and inert gases (Neon, Argon).
- **Principle:** Accelerated electrons cause mercury molecules to produce UV radiation. A phosphor coating on the glass fluoresces to produce visible white light.
- **Component:** A choke (inductor) is used to start the lamp and regulate current.
- **Efficiency:** Converts about 22% of electrical energy into light.

Fluorescent Lamps**Light Emitting Diodes (LEDs)**

- **Principle:** Electroluminescence (light emitted when electrons recombine with holes in a semiconductor like Gallium Arsenide).
- **Advantages of CFLs:**
 - More energy efficient.
 - Longer lifespan (~50,000 hours vs. CFL's 6,000-10,000 hours).

Light Emitting Diodes (LEDs)

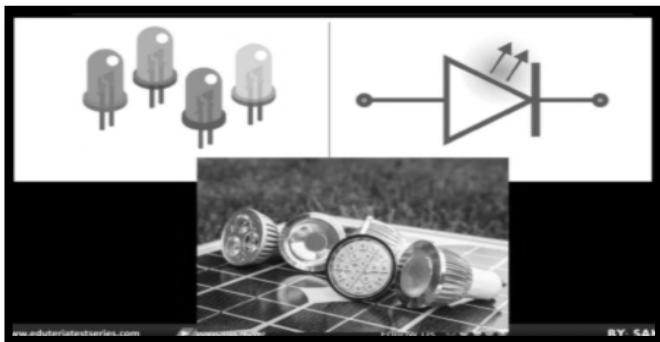
Advantages over Sodium Vapor Lamps (for street lighting):

- Directional light emission (~180 degrees), reducing waste.
- Longer lifespan.
- Better color rendering (not monochromatic yellow).





LED



Sodium Vapor Lamps

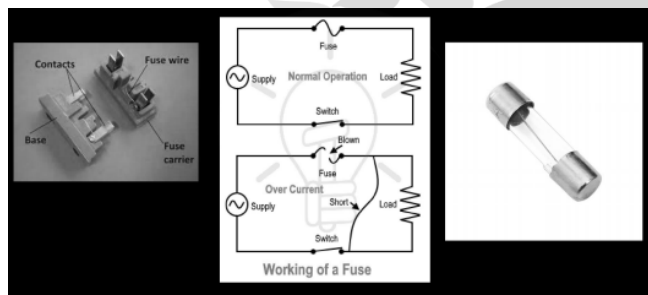
- Produces light in all directions (360 degrees).
- Emits an almost monochromatic yellow light.

Safety Devices and Domestic Systems

Safety Devices

- **Principle:** Works on the heating effect of electricity.
- **Function:** Safety device that melts and breaks the circuit during overcurrent.
- **Characteristics:** High resistance, low melting point, made from tin, lead, or zinc alloys.

FUSE

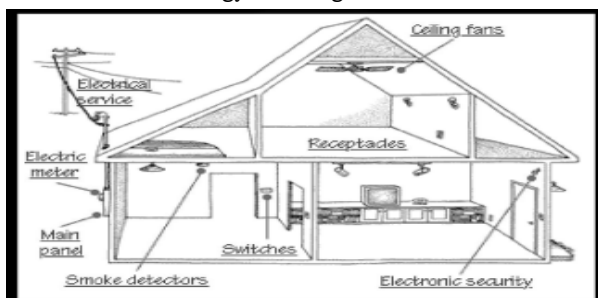


Earthing (Grounding)

- Provides a safe path for fault current to flow into the ground.
- In a three-pin plug, the longest pin is the Earth terminal (connects first, disconnects last).

Lightning Conductor

- A metal rod that safely conducts lightning's electric energy to the ground.



Domestic Electrical Systems

- **Wiring Configuration:** Basically, a parallel connection.
- **Wire Color Coding (Three-wire system):**
 - **Live Wire:** Red insulation.
 - **Neutral Wire:** Black insulation.
 - **Earth Wire:** Green insulation.

Faraday Constant (F):

- A universal constant representing the charge carried by one mole of electrons.
- Value: ~96,485 Coulombs per mole (C/mol).

Current Flow

- Current does not flow between two charged bodies if they are at the same electric potential. A potential difference is required.

Power Consumption of Appliances (Ascending Order)

- Fan < Television < Electronic Press (Iron) < Electronic Kettle.



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